

FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-111 Capstone Data & Source Contracts

RetailPulse grain, keys, change semantics, schema policy, quality and reconciliation

Status	RC1 - FINAL-LAYER QA PASS CANDIDATE
Course boundary	C3-104 = C3-09 phase Gate only; C3-113 = separate course-wide graduation Gate
Release boundary	Course 3 is NOT FINAL until Mentor/PWA + whole-course release regression are complete.

Permanent engineering behavior controls the course. Vendor UI/version notes are refreshable references, and external links are support-only; the course artifacts remain sufficient without them.

1. Canonical source registry

orders	Silver current: one row/order_id	order_id	updated_at correction + insert	required fields; current-key uniqueness; amount >= 0; status domain; customer FK
customers	one row/customer_id	customer_id	snapshot/replace	unique key; required ID; declared region/segment rules

2. Order field contract

order_id	Stable nonblank business key; exactly one trusted current row per key.
customer_id	Required FK; unresolved key is blocked/quarantined according to explicit policy.
order_date	Business event date for serving; not the change-order timestamp.
status	Baseline domain: Completed, Pending, Cancelled. New values require explicit contract/business-rule change.
amount	Numeric, non-negative; contributes to completed-revenue control when status=Completed.
updated_at	Change-order field; newer valid version wins. Late corrections must remain discoverable.

3. Baseline invariants

Base rows	5
Incremental rows	3
Current rows	7
O102 amount	1050.00
O106 / O107	once each
Completed revenue	6910.00
2026-09-01 completed	2 orders / 2300.00
2026-09-02 completed	1 order / 1620.00
2026-09-03 completed	2 orders / 2990.00

4. Schema-change policy

Add nullable field	Update contract + compatibility test before serving use.
Rename/remove field	Breaking unless producer/consumer migration is versioned and validated.
Type change	Breaking unless compatibility is explicitly proven with representative data.
New enum/status	Block/quarantine until business rule and downstream behavior are updated.
Privacy classification changes	Update access/evidence policy before exposure.

5. Reconciliation contract

- Source-current order key set equals trusted Silver/serving current-state key set.
- Trusted current state has no duplicate order_id.
- Critical counts and completed revenue reconcile after clean run, recovery, backfill and controlled change.
- Rejected/quarantined rows are counted, reason-coded and excluded from trusted publish until resolved.
- Any unexplained mismatch blocks trusted publish; a green scheduler/run status never overrides failed reconciliation.

6. Contract version evidence

For every controlled change save: change request → affected contract/version → compatibility classification → tests → implementation → post-change reconciliation → consumer/runbook note. Do not edit a contract silently after the data already changed.

7. Fresh-change examples C3-113 may use

- Nullable sales_channel field.
- New status/domain value requiring explicit approval.
- Freshness SLO change.
- New consumer access boundary.
- Serving calculation change with backward-compatibility impact.